

Bryce Grant | EE PhD Candidate

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EDUCATION & HONORS

Case Western Reserve University

Cleveland, OH

PhD Candidate (Robotic Perception)

Expected: May 2028

Graduate Advisor: [Dr. Peng \(Edward\) Wang](#)

Graduate Honors: NSF Graduate Research Fellow (GRFP 2025), Advanced to Candidacy, NVIDIA Academic Grant

University of Kentucky: Magna Cum Laude, Lexmark Scholar

Lexington, KY

Dual B.S. in Computer Engineering and Electrical Engineering (CS, Math Minors)

Graduated: May 2024

PUBLICATIONS & PREPRINTS

SPARK: Sequential Planning via Anchored Robotic Keypoints

[B. Grant](#), A. Rothenberg*, L. Senning*, P. Wang

Under Review | [Website](#)

TrianguLang: Geometry-Aware Semantic Consensus for Pose-Free 3D Localization

[B. Grant](#), A. Rothenburg, A. Banerjee, P. Wang

Under Review | [Website](#)

Not All Features Are Created Equal: A Mechanistic Study of Vision-Language-Action Models

[B. Grant*](#), X. Zhao*, P. Wang

Under Review | *ICLR MM Intelligence Workshop (Oral)*, 2026. [Website](#)

Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representation

[B. Grant](#) and P. Wang

Under Review | *ICLR UCRL Workshop*, 2026. [Website](#)

Quaternion Approximate Networks for Enhanced Image Classification and Oriented Object Detection

[B. Grant](#) and P. Wang

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS Oral), 2025. [Website](#)

EXPERIENCE

Case Western Reserve University

Cleveland, OH

Graduate Research Assistant

Aug. 2024 – Present

- Built SPARK, a training-free robotic manipulation system that more than doubled the best VLA and code-generation baselines on out-of-distribution tasks (LIBERO-Pro), deployed zero-shot across three real robot platforms (UR10e, Franka FR3, bimanual Franka)
- Co-led an early large-scale mechanistic interpretability study of VLA models (6 models, 424 sparse autoencoders); released a ~7TB dataset and maintain Action Atlas
- Leading 3 undergraduate researchers on robotic perception and manipulation projects, guiding project scoping, code reviews, and experiment design

Mercor

Remote

Applied AI Engineer Internship

Aug. 2025 – Apr. 2026

- Led internal data analytics team to track project development, individual contractor productivity, task accuracy, and budget allocation across multi-million dollar RLHF and SFT data collection operations
- Built a video QA pipeline for egocentric manipulation datasets, automating multi-level validation to ensure data readiness for sim-to-real transfer
- Scaled Mercor's code data pipeline from \$0 to 8-figure revenue, partnering with leading AI labs to generate high-quality post-training and evaluation datasets for coding agents; architected and owned all engineering systems from data ingestion to final output
- Led development of reinforcement-learning environments for code, designing next-generation benchmarks to evaluate AI coding agents on tool use, contextual reasoning, and multi-step problem solving

HP

Vancouver, WA

PHD ML Intern

May 2024 – Aug. 2024

- Deployed an anomaly detection pipeline on AWS for enterprise cloud spend analysis, combining LSTM forecasting with STL decomposition to flag cost anomalies across accounts
- Engineered a comparative RAG framework: benchmarked semantic vs. recursive chunking, BM25 vs. FAISS retrieval, and LLM query expansion strategies, with evaluation via coherence and NDCG metrics
- Built a customer-facing LLM chat agent that guided users through printer setup, replacing static manuals with dynamic reranking and multi-stage error recovery

University of Kentucky

Lexington, KY

AI for Smart Manufacturing Lab Research ^{COPI}Aug

Aug. 2023 – May. 2024

- Developed CNN-Transformer hybrid for 6-DOF pose estimation from RGB-D inputs for robotic arm manipulation
- Led navigation subsystem for autonomous telepresence robot, implementing 3D SLAM with servo-mounted LiDAR, achieving real-time mapping and dynamic obstacle avoidance in dynamic environments

Honeywell
Embedded Software Engineer Intern

Atlanta, GA
May 2023 – Aug. 2023

- Developed bare-metal firmware API for life safety microcontrollers, supporting product line with 100K+ unit volume
- Validated dual-CPU communication protocols using oscilloscope debugging and hardware-in-the-loop testing

SKILLS

Programming Languages: C/C++, CUDA, MATLAB, Python, SQL

ML & Robotics: COLMAP, Distributed Training, HuggingFace, Isaac Lab, JAX, LiDAR, MuJoCo, OpenCV, Open3D, Pytorch, RL, ROS2, WandB, VLAs, World Models

Cloud, DevOps & Engineering: Blender, Docker, Linux, PySpark, Snowflake

Domains & Methodologies: Benchmark Engineering, Computer Vision, Geometric Deep Learning, LLM Evaluation, Mechanistic Interpretability, RL, Robotics

Robotic Platforms: Franka Research 3, Unitree (G1 and Go2), UR (10e, 5e)

SERVICE

Conference Reviewer

Dec.2024 – Present

- CoRL 2026, ECCV 2026, ICLR 2026, IROS 2025-2026, NAMRC 25-26

Mathworks Student Ambassador

Oct. 2024 – Dec. 2025

- Hosted Hackathons and educational events while increasing MATLAB brand awareness across campus

NSBE

Lexington, KY

Region III Finance Chair

May 2023 – May 2024

- Orchestrated a regional career fair with 50+ participating companies while raising over \$250,000 in revenue in six months